Laboratory Record

of _____

Roll No.: _____

Experiment No.: _____

Sheet No.: _____

Date.: _____

Aim

To design and implement an interactive, web-based calculator capable of performing basic arithmetic operations using HTML, CSS, and JavaScript.

Description

This project is a simple client-side calculator application. The user interface is structured with HTML using an input field for the display and a grid of buttons for numbers and operators. CSS is utilized to style the calculator, providing a clean, modern look using CSS Grid for the layout and adding interactive visual feedback (hover and active states) for the buttons.

The core logic is driven by JavaScript, which captures user interactions, manages the state of the first and second operands, and performs mathematical operations (addition, subtraction, multiplication, and division). It also includes basic error handling, such as preventing calculations from dividing by zero.

Algorithm

1. Initialize: Create empty variables to track num1, num2, and the operator.
2. Input: Append clicked numbers to the display screen.
3. Operator: On operator click (+, -, *, /), save the display to num1, store the operator, and clear the screen.
4. Calculate: On = click, save the current display to num2.
5. Evaluate: Perform the math operation on num1 and num2, then display the result.
6. Clear: On C click, reset the display and all variables.

Program Code:

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Calculator</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="calculator">
    <input type="text" id="display" readonly>
```

```
    <div class="buttons">
        <button onclick="clearDisplay()">C</button>
        <button onclick="add('/')">/</button>
        <button onclick="add('*')">*</button>
        <button onclick="add('-')">-</button>
        <button onclick="add('7')">7</button>
        <button onclick="add('8')">8</button>
        <button onclick="add('9')">9</button>
        <button onclick="add('+')">+</button>
        <button onclick="add('4')">4</button>
        <button onclick="add('5')">5</button>
        <button onclick="add('6')">6</button>
        <button onclick="calculate()">=</button>
        <button onclick="add('1')">1</button>
        <button onclick="add('2')">2</button>
        <button onclick="add('3')">3</button>
        <button onclick="add('0')">0</button>
    </div>
  </div>
  <script src="script.js"></script>
</body>
</html>
```

Script.js

```
let num1 = "";
let num2 = "";
let operator = "";
const display = document.getElementById("display");
const buttons = document.querySelectorAll("button");
function add(value) {
```

```
    buttons.forEach(button => button.classList.remove("active"));
    event.target.classList.add("active");
    if (value === "+" || value === "-" || value === "*" || value === "/") {
        num1 = display.value;
        operator = value;
        display.value = "";
    } else {
        display.value += value;
    }
}
function calculate() {
    buttons.forEach(button => button.classList.remove("active"));
    event.target.classList.add("active");
    if (num1 === "" || operator === "" || display.value === "") {
        return;
    }
    num2 = display.value;
    num1 = Number(num1);
    num2 = Number(num2);
    if (operator === "+") {
        display.value = num1 + num2;
    }
    else if (operator === "-") {
        display.value = num1 - num2;
    }
    else if (operator === "*") {
        display.value = num1 * num2;
    }
    else if (operator === "/") {
```

```
        display.value = num2 === 0 ? "Error" : num1 / num2;
    }
    num1 = "";
    num2 = "";
    operator = "";
}
function clearDisplay() {
    buttons.forEach(button => button.classList.remove("active"));
    display.value = "";
    num1 = "";
    num2 = "";
    operator = "";
}
```

Style.css

```
body {
    font-family: Arial;
    background: #f5f5f5;
}
.calculator {
    width: 250px;
    margin: 100px auto;
    padding: 15px;
    background: white;
    border: 1px solid #ddd;
}
#display {
    width: 100%;
    box-sizing: border-box;
    padding: 10px;
```

```
    margin-bottom: 10px;
    font-size: 20px;
}
.buttons {
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 5px;
}
button {
    padding: 15px;
    font-size: 16px;
    border: 1px solid #ccc;
    background: white;
}
button:hover {
    background: #eee;
}
button.active {
    background: #ddd;
}
```





